

# MAT 017B Quiz 6

August 28, 2018

Name: Key

SID: \_\_\_\_\_

- Denote by  $L(t)$  the length of a certain fish at time  $t$ , and assume that this fish grows according to the von Bertalanffy equation

$$\frac{dL}{dt} = k(L_{\infty} - L(t)) \quad \text{with } L(0) = 5$$

where  $k$  and  $L_{\infty}$  are positive constants. A study showed that the asymptotic length is equal to 150 in. and that it takes this fish 40 months to reach a third of its asymptotic length. Use this information to determine the constants  $k$  and  $L_{\infty}$ .

$$\frac{dL}{dt} = k(L_{\infty} - L)$$

$$\int \frac{dL}{L_{\infty} - L} = \int k dt \quad -\ln|L_{\infty} - L| = kt + C$$

$$L_{\infty} - L = Ce^{-kt}$$

$$L(t) = L_{\infty} - Ce^{-kt}$$

$$\lim_{t \rightarrow \infty} L(t) = \lim_{t \rightarrow \infty} L_{\infty} - \cancel{Ce^{-kt}}^0 = 150$$

$$\boxed{L_{\infty} = 150}$$

$$L(0) = 150 - Ce^{-k \cdot 0} = 150 - C = 5 \Rightarrow C = 145$$

$$\frac{1}{3}150 = 150 - 145e^{-40k}$$

$$\frac{-100}{-145} = e^{-40k} \Rightarrow$$

$$\boxed{k = -\frac{1}{40} \ln\left(\frac{100}{145}\right)}$$